Executive Summary: Asset Reliability and Risk

Where to focus maintenance, technician and spare-parts effort to cut downtime and production loss

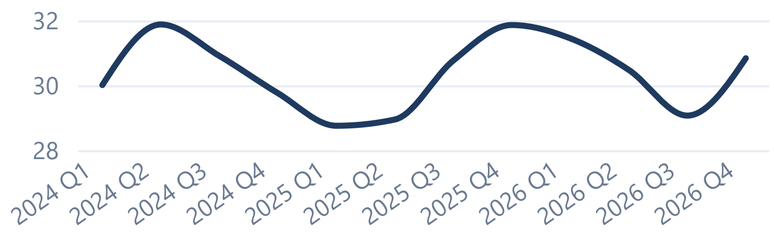
Year: All | Asset class: All | Criticality: All | Maintenance strategy: All

Showing: all asset classes | all criticality levels | all strategies | 2024-2026 | 5,000 records

Production loss is 4.5x the repair spend. Highest exposure: Railway Signalling at $55,991 loss per record. Availability is 97.0%, but MTBF is only 313 h.

Availability	MTBF (h)	MTTR (h)	Production Loss	Loss to Maintenance Ratic	Watchlist Records
97.0%	313.3	12.0	$229M	4.5x	1,418
= portfolio	= portfolio	= portfolio	100.0% of portfolio loss	= portfolio	28.4% of records

Downtime per 1,000 planned hours



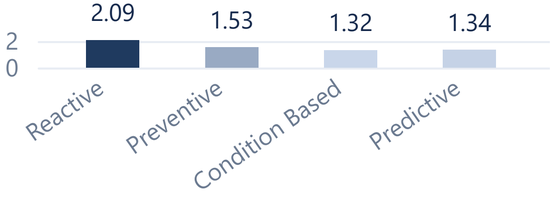
Production loss per record by asset class (USD)



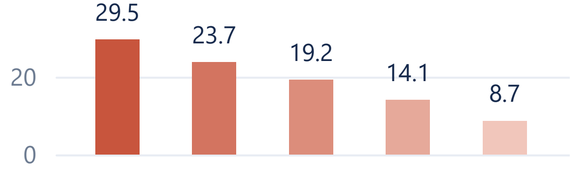
Loss per record: criticality x strategy (USD)

Criticality	Reactive	Preventive	Condition Based	Predictive
Low	$62,325	$44,789	$32,117	$36,356
Medium	$59,504	$45,699	$36,935	$37,250
Total	**$62,030**	**$46,658**	**$39,543**	**$40,216**

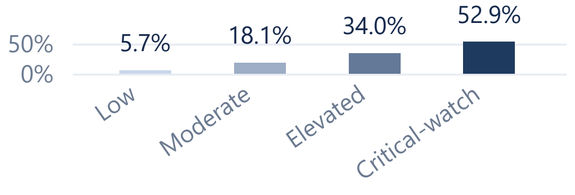
Failures per record by strategy



Downtime per record by condition score



Actual high-loss rate by risk tier



Decisions requested

1 Review Critical-asset restoration: MTTR 16.3 h and only 29% of records meet SLA.
2 Pilot condition-based care on ageing, degraded assets.
3 Trial the risk watchlist as a ranking aid, with human approval.
4 Fix data definitions before roll-out.

KPI Scorecard

Every card shows the KPI, its quarterly trend and how the current selection compares with the full portfo

Year: All

Asset class: All

Criticality: All

Maintenance strategy: All

Showing: all asset classes | all criticality levels | all strategies | 2024-2026 | 5,000 records

RELIABILITY

Availability	Utilization	MTBF (h)	MTTR (h)	Downtime per 1000 Plann	Failures per Record
97.0%	79.4%	313.3	12.0	30.4	1.52
= portfolio	= portfolio	= portfolio	= portfolio	= portfolio	= portfolio

COST AND CONSEQUENCE

Maintenance Cost	Production Loss	Economic Exposure	Cost per Operating Hour	Downtime Cost Rate	Loss to Maintenance Ratic
$50M	$229M	$279M	$21.19	$2,513	4.5x
100.0% of portfolio spend	100.0% of portfolio loss	100.0% of portfolio exposu	= portfolio	= portfolio	= portfolio

EXECUTION AND RISK

Overdue Rate	First-Time Fix	Repeat Repair	SLA On-Target	Watchlist Records	High-Risk Records
14.1%	76.6%	23.4%	46.1%	1,418	289
= portfolio	= portfolio	= portfolio	= portfolio	28.4% of records	5.8% of records

Reliability and Failure Intelligence

Which assets and conditions drive failures and downtime? MTBF = operating hours / failures; MTTR = d

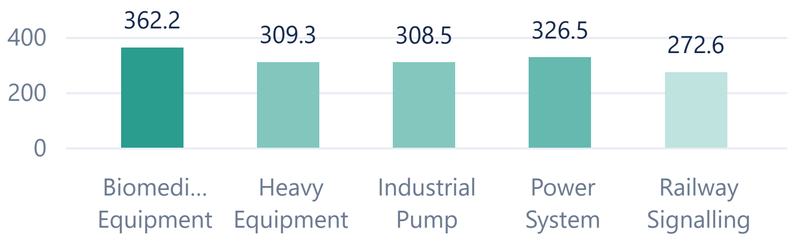
Year: All | Asset class: All | Criticality: All | Maintenance strategy: All

Showing: all asset classes | all criticality levels | all strategies | 2024-2026 | 5,000 records

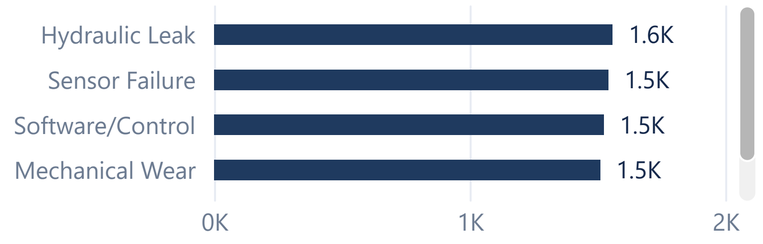
MTBF 313 h and MTTR 12.0 h. Weakest class: Railway Signalling (273 h MTBF). Failures rise with age and falling condition score.

MTBF (h)	MTTR (h)	Availability	Failures per Record	Downtime per Record	Utilization
313.3	12.0	97.0%	1.52	18.2	79.4%
= portfolio	= portfolio	= portfolio	= portfolio	= portfolio	= portfolio

MTBF (h) by asset class



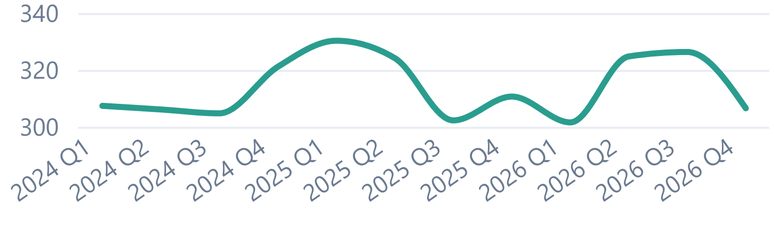
Failures by failure mode



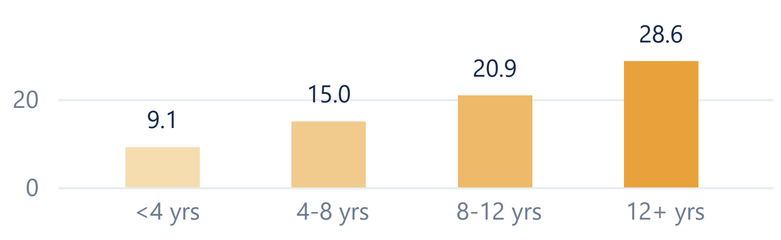
Failures per record: class x strategy

Asset_Class	Reactive	Preventive	Condition Basec
Biomedical Equipment	1.96	1.27	1.09
Heavy Equipment	2.10	1.56	1.33
Total	**2.09**	**1.53**	**1.32**

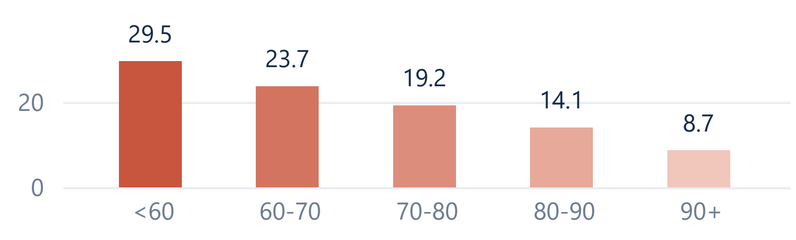
MTBF (h) by quarter



Downtime per record by age band



Downtime per record by condition score



Work-Order Performance

Overdue = overdue WOs / (preventive + corrective); first-time fix = FTF / corrective; SLA on-target = re

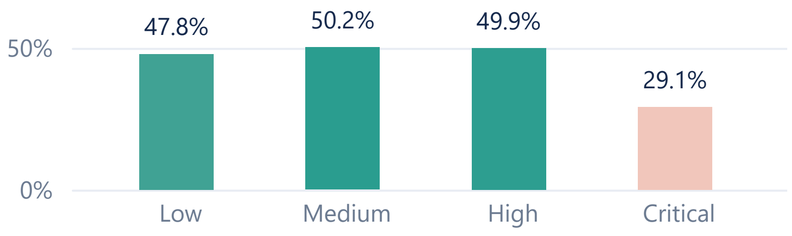
Year: All | Asset class: All | Criticality: All | Maintenance strategy: All

Showing: all asset classes | all criticality levels | all strategies | 2024-2026 | 5,000 records

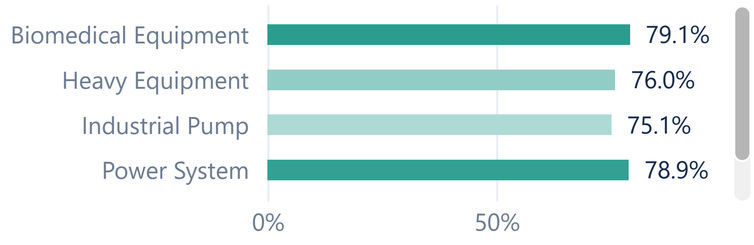
Overdue rate 14.1%, first-time fix 76.6%, repeat repair 23.4%. Only 46% of records meet their SLA target; Critical assets: 29%.

Overdue Rate	First-Time Fix	Repeat Repair	SLA On-Target	Preventive Share	Overdue WOs
14.1%	76.6%	23.4%	46.1%	75.2%	5,940
= portfolio	= portfolio	= portfolio	= portfolio	= portfolio	100.0% of portfolio overdu

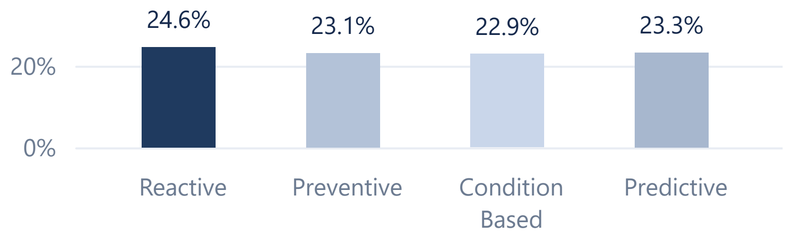
SLA on-target share by criticality



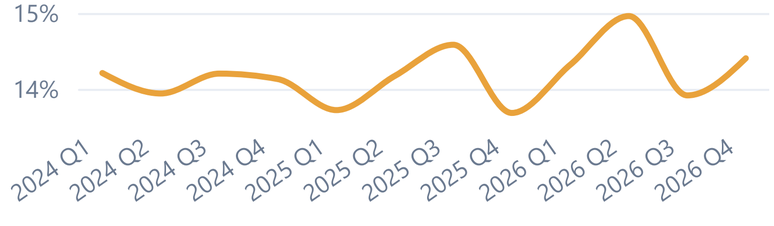
First-time fix by asset class



Repeat repair rate by strategy



Overdue rate by quarter



First-time fix: team x shift

Technician_Team	Day	Night	Total
Team A	74.3%	78.5%	**76.3%**
Team B	76.4%	76.9%	**76.6%**
Total	**76.2%**	**77.0%**	**76.6%**

Overdue rate: team x shift

Technician_Team	Day	Night	Total
Team A	13.7%	14.4%	**14.0%**
Team B	14.2%	14.7%	**14.4%**
Total	**14.0%**	**14.3%**	**14.1%**

Cost and Production-Loss Intelligence

Maintenance cost understates failure consequences: production loss is 4.5x repair spend. "Other" = mai

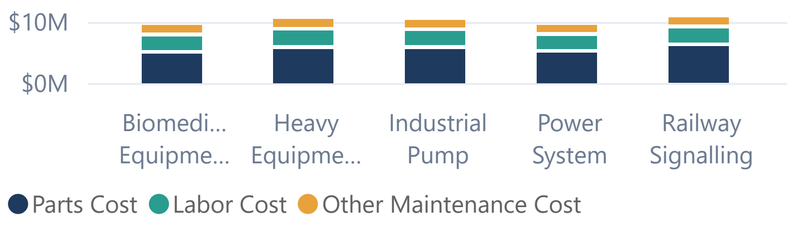
Year: All | Asset class: All | Criticality: All | Maintenance strategy: All

Showing: all asset classes | all criticality levels | all strategies | 2024-2026 | 5,000 records

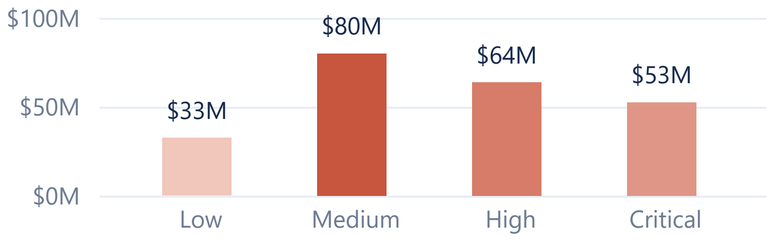
Economic exposure $279.3M: $50.4M repair spend and $228.9M production loss. Every downtime hour costs $2,513.

Maintenance Cost	Production Loss	Economic Exposure	Cost per Operating Hour	Downtime Cost Rate	Loss per Record
$50M	$229M	$279M	$21.19	$2,513	$45,776
100.0% of portfolio spend	100.0% of portfolio loss	100.0% of portfolio exposu	= portfolio	= portfolio	= portfolio

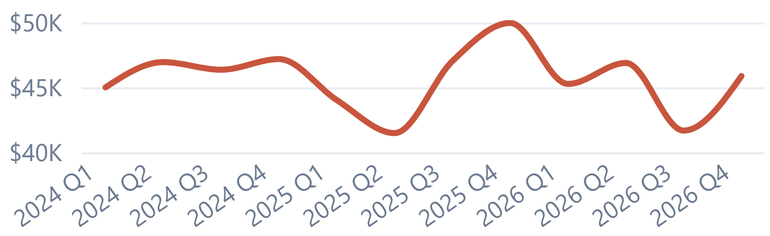
Maintenance cost components by asset class



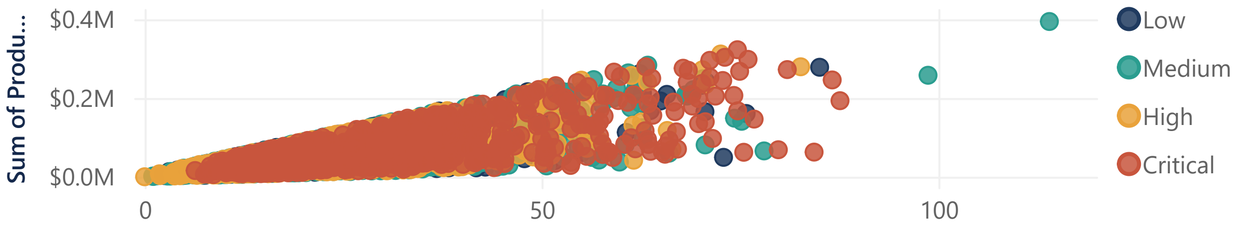
Production loss by criticality



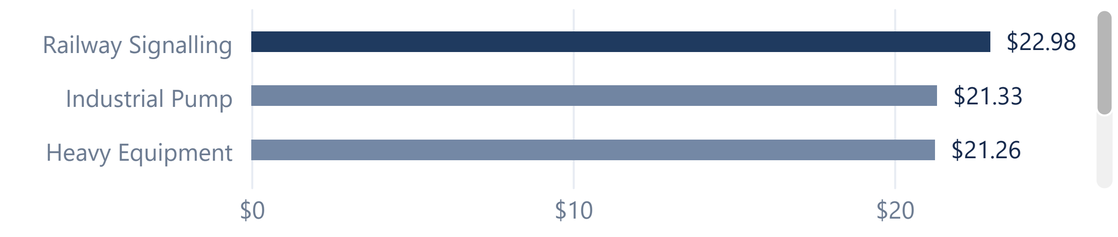
Production loss per record by quarter



Downtime hours vs production loss (each point = one record)



Cost per operating hour by asset class (USD)



Predictive Risk and Intervention Priority

High-loss = top 20% production-loss record. Logistic model, chronological test AUC 0.73 vs 0.61 baseli

Year: All

Asset class: All

Criticality: All

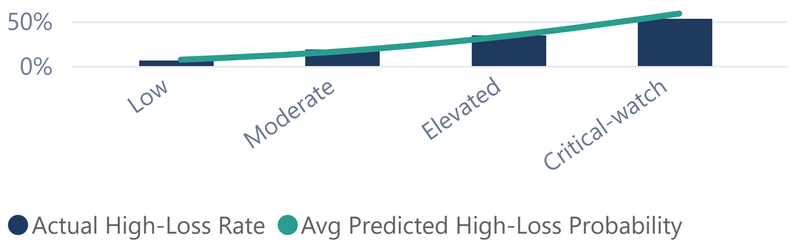
Maintenance strategy: All

Showing: all asset classes | all criticality levels | all strategies | 2024-2026 | 5,000 records

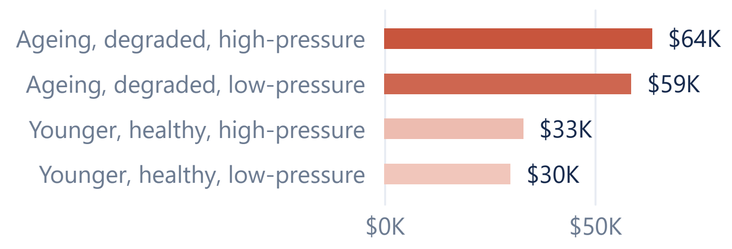
Watchlist: 1,418 records hold 44% of production loss. Critical-watch tier: 53% high-loss rate vs 20% overall. Ranking aid only: human approval required.

Watchlist Records	Watchlist Loss Share	Actual High-Loss Rate	Avg Predicted High-Loss F	High-Risk Records	Critical-watch High-Loss R
1,418	43.6%	20.3%	20.1%	289	52.9%
28.4% of records	+15.2 pts above its record s	= portfolio	-0.2 pts vs actual	5.8% of records	+32.6 pts above overall

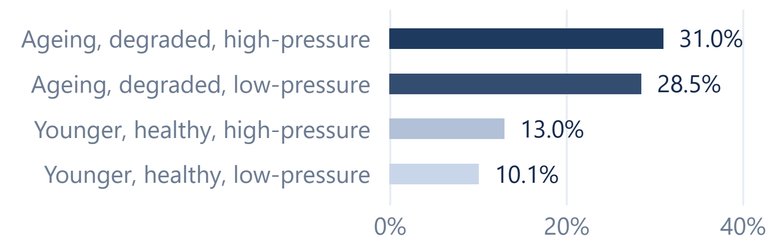
Calibration: actual vs predicted high-loss rate



Loss per record by condition profile (USD)



Actual high-loss rate by condition profile



Exception queue: Critical-watch tier, highest predicted probability first

Record_ID	Record_Date	Asset_Class	Criticality	Maintenance_Strategy	Condition_Score	Vibration_mm_s	Asset_Age_Years	HighLoss_Score	Production_Loss_USD
T3-00329	2026-05-20	Railway Signalling	Critical	Preventive	49.87	6.20	17.58	81.4%	$34,288
T3-02950	2024-10-25	Biomedical Equipment	Critical	Reactive	55.18	4.66	12.79	77.8%	$70,971
T3-04591	2024-07-05	Railway Signalling	Critical	Reactive	60.41	5.25	15.17	76.6%	$99,019

Method, Definitions and Data Governance

How to read the numbers and what they can and cannot support

KPI definitions (weighted: sum / sum)

Availability = 1 - sum(Downtime h) / sum(Planned h)
Utilization = sum(Operating h) / sum(Planned h)
MTBF = sum(Operating h) / sum(Failures); BLANK if no failures
MTTR = sum(Downtime h) / sum(Failures)
Cost per operating hour = sum(Maintenance cost) / sum(Operating h)
Downtime cost rate = sum(Production loss) / sum(Downtime h)
Preventive share = sum(Preventive WO) / (sum(Preventive) + sum(Corrective))
Overdue rate = sum(Overdue WO) / (sum(Preventive) + sum(Corrective))
First-time fix = sum(FTF count) / sum(Corrective WO)
Repeat repair = sum(Repeat count) / sum(Corrective WO)
SLA on-target = share of records with actual <= target hours
Comparison chips = selection minus full portfolio (same measure, all filters removed).
Why not row means: averaging row rates ignores exposure weights. Example: MTBF 313 h weighted vs 343 h row mean.

Data and governance

Source: Group 9226 Team 3 Enterprise Dataset v1.0 (synthetic educational panel), 5,000 records x 60 variables, 2024-01-05 to 2026-12-25. Master unchanged; data embedded in this file (no external path).
Added fields: High_Loss_Flag, SLA_On_Target, Other_Maint_Cost_USD, age and condition bands, model score, risk tier, condition profile.
Checks: 0 missing, 0 duplicates; availability, utilization, MTBF, loss and work-order identities hold.
Flags: Asset_ID, Site, Region and Criticality are independent row labels (every ID appears under all 5 classes): no unique-asset counts or per-asset trends. 166 rows are future-dated and 2026 holds about half the records of 2024/2025, so compare years by rate. Maintenance cost is not parts + labor (about 18% other). Only 3 safety incidents and 84 stockouts.
Findings are associations in synthetic data, not causal effects or realised savings.

AI layer and its limits

Model: logistic regression for a high-loss record (top 20% of production loss; threshold fixed on the training period). Pre-decision inputs only: class, criticality, strategy, age, planned hours, temperature, vibration, pressure, condition score, energy, parts lead days, SLA target.
Validation: chronological. Train 2024-01 to 2025-08 (3,340 rows), test 2025-09 to 2026-12 (1,660 rows).
Result: test AUC 0.73 (95% CI 0.70 to 0.76) vs 0.61 for a class x criticality baseline; top-decile lift 2.5x. Gradient boosting did not beat it.
Limits: the supplied Maintenance_Priority_Score ranks about as well (AUC 0.73); Failure_Probability is a near-formula of the inputs (R2 0.998), so it was not a feature.
Scores are out-of-sample (test) or out-of-fold (training). Tiers: Low <10%, Moderate 10-25%, Elevated 25-50%, Critical-watch >50%.
Predictive performance is not achieved savings. AI assistance is disclosed in the report.

Asset Class Detail (drill-through)

Right-click an asset class on any page and choose Drill through

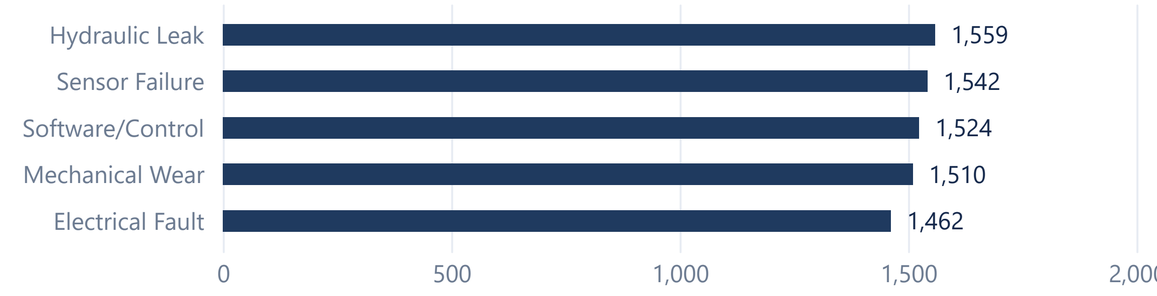
Back

Records	Availability	MTBF (h)	MTTR (h)	Loss per Record	Watchlist Records
5,000	97.0%	313.3	12.0	$45,776	1,418
100.0% of records	= portfolio	= portfolio	= portfolio	= portfolio	28.4% of records

Downtime per 1,000 planned hours by quarter



Failures by failure mode



Records of the selected asset class, by production loss (highest first)

Record_ID	Record_Date	Criticality	Maintenance_Strategy	Failure_Mode	Condition_Score	Downtime_Hours	Production_Loss_USD	HighLoss_Score	Priority_Tier
T3-02114	2025-02-20	Medium	Preventive	Hydraulic Leak	61.32	113.98	$394,436	42.2%	Elevated
T3-04342	2025-10-10	Critical	Condition Based	Hydraulic Leak	66.68	74.68	$322,764	32.1%	Elevated
T3-01099	2025-07-20	High	Preventive	Hydraulic Leak	55.91	72.57	$311,888	69.3%	Critical-watch
T3-04354	2024-10-20	Critical	Reactive	Electrical Fault	78.95	73.09	$303,492	73.8%	Critical-watch
T3-03684	2025-12-20	Critical	Predictive	Sensor Failure	93.50	76.02	$298,253	19.0%	Moderate
T3-00992	2024-08-10	Critical	Preventive	Mechanical Wear	72.76	71.20	$295,772	23.1%	Moderate
T3-04978	2026-10-15	Medium	Preventive	Hydraulic Leak	78.78	63.39	$283,510	16.6%	Moderate
T3-01519	2025-07-20	Critical	Reactive	Hydraulic Leak	61.72	63.13	$279,504	57.7%	Critical-watch